Hyperspectral Target Detection via Spectral Aggregation and Separation Network With Target Band Random Mask

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***Abstract*— Hyperspectral target detection (HTD) is a pixel-wise detection method based on limited prior targets and spectral differences, which has been widely studied and applied in many fields. Recently, deep learning (DL) plays an important role in hyperspectral imagery (HSI) processing. However, for HTD, the severe lack of class-balanced training sets is an enormous challenge. Meanwhile, it is difficult to suppress backgrounds while highlighting targets through the deep network. To address these issues, we propose a spectral aggregation and separation network (SASN) with a target band random mask (TBRM) for HTD in this article. For the training sets of SASN, a multifarious representative background selection strategy (MRBS) is first proposed to obtain a multifarious and representative background training set. Next, aiming at the notorious class imbalance, a data augmentation (DA) method, TBRM, is proposed to generate adequate target training set by repeating randomly zero-masking the spectral bands of a prior target. Subsequently, in the training of SASN, residual connection and squeeze-and-excitation (SE) channel attention mechanism are applied to fully extract high discriminative features and nonlinear ones in the spectra. Besides, to better separate the targets and backgrounds, a triplet-soft loss function is presented, which makes the training in the direction of spectral separation of background samples from both the prior target and target samples. During testing, the trained SASN distinguishes the spectral similarities and differences simultaneously for highlighting targets and suppressing backgrounds. Moreover, extensive experimental results validate that the proposed method has superior detection performances, background suppression capacity, and separability compared with ten cutting-edge HTD algorithms on six benchmark HSI datasets.**

***Index Terms*— Data augmentation (DA), deep learning (DL), hyperspectral imagery (HSI), hyperspectral target detection (HTD).**

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I. Introduction

HYPERSPECTRAL imagery (HSI) is 3-D with abundant spectral information acquired by imaging spectrometers, and its spectral bands can describe the information about ground coverings well [1], [2]. Hyperspectral target detection (HTD), one of the well-known HSI processing applications, has been extensively studied in recent decades [3], [4]. Depending on limited prior targets, HTD is usually a pixel-wise detection by distinguishing the subtle spectral differences between backgrounds and targets. Currently, HTD plays an important role in many fields, such as agriculture [5], ground object observation, military reconnaissance [6], and mineral exploration [7].

Since there are only positive (target) and negative (background) samples, HTD is usually considered a binary classification problem [8]. Besides, due to the lack of prior knowledge of the datasets, most HTD algorithms are designed in terms of known targets and unknown backgrounds. Generally, the detection result of each pixel obtained by the HTD algorithm is a fraction. The closer the fraction is to 1, the higher the probability that the pixel belongs to the target category. On the contrary, the closer the fraction is to 0, the more likely it is that the pixel is the background category. Consequently, the design goal of the HTD algorithms is to make the target pixels closer to the prior target and the distance between the background pixels and the prior target larger. Notably, the target pixels usually occupy very small and sparse parts of the HSI. In addition, the background pixels usually include many complex and unknown types of ground material and some interfering substances different from ordinary noise [9]. Therefore, the key of HTD is to locate, separate, and highlight the target pixels from vast background pixels simultaneously suppressing the background pixels.

Since the 1990s, plentiful traditional signal processing-based HTD algorithms have been developed. Most fundamentally, the Euclidean distance (ED) [10], spectral angle mapping (SAM) [11], and spectral information divergence (SID) [12] were used to measure the spectral differences

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thereupon valid detection effect is achieved. Subsequently, numerous HTD algorithms based on statistics and assumptions were further proposed. Based on the assumption of multivariate normal distribution, the generalized likelihood ratio test (GLRT) [13] and its adaptive version, adaptive cosine/coherent estimator (ACE) [14], perform target detection based on the linear mixing model (LMM) [15], [16], [17]. Moreover, spectral match filter (MF) [18] and its derivative approaches were proposed, such as adjusted MF (AMF) [19] and regularized MF (RMF) [20]. Furthermore, constrained energy minimization (CEM) [21], [22], [23] achieves excellent performances by amplifying targets and suppressing the backgrounds under constrained conditions. Inspired by CEM, kernel CEM (KCEM) [24], hierarchical CEM (HCEM) [25], and diverse-direction CEM (D^2CEM) [26] constantly sprang up to improve detection performance from different aspects. Although the abovementioned classic detectors are generic and excellent, the characteristics of real HSI, such as spectral variability and subpixel interference may result in degraded performances [27], [28].

Moreover, considering the sparsity, low-rank attributes [29], [30], [31], and the possible nonlinearity [32] of HSI, abundant machine learning (ML)-based HTD algorithms have been investigated. The renowned sparse representation target detection (SRTD) [33] and combined sparse and collaborative representation detector (CSCR) [34] construct the local background dictionaries by the dual concentric window, yielding good results. Besides, Zhu et al. [35] proposed a single-spectrum-driven binary-class sparse representation target detector (SSBSTD), which achieves promising detection results by calculating the sparse coefficients of targets and backgrounds, respectively, and only needs one prior target. Furthermore, considering the features of the data itself, Sun et al. [36] proposed a tree-structured encoding-based detector (TSETD) for HTD to separate the backgrounds and targets. The aforementioned ML-based HTD algorithms attach more significance to the characteristics of real HSI when designing, yielding better performances. Nevertheless, they generally require manual participation in feature selection and extraction, which makes features lack expressiveness.

In recent years, end-to-end deep learning (DL) without expert experience and knowledge provides new ideas for HTD, owing to its powerful ability to automatically mine the intrinsic, connotative, and potentially valuable features from the massive data [37], [38]. However, the extremely unbalanced data cannot provide reasonable training set for DL-based HTD. In the initial studies, Du et al. [39] proposed a convolutional neural network (CNN)-based HTD (CNNTD), which needs manually re-marking additional abundant labels to construct a sufficient training set. Intelligently, Feng et al. [40] proposed a spectral–spatial joint HTD (SSJTD) to address this issue through transfer learning, yet the datasets used usually require high costs and are difficult to obtain. On this basis, Zhang et al. [41] proposed a deep CNN for HTD (HTD-Net), the input of which is the spectral differences of pair pixels. Therein the data preparation by an autoencoder-based U-Net (U-AE) and the criterion of linear prediction are innovative but it is cumbersome and not entirely reasonable. In addition to dealing with the problem of class imbalance, scholars pay more and more attention to the separation of targets and backgrounds in the detection results. For instance, Zhu et al. [42] proposed a two-stream CNNTD (TSCNTD), which is in the light of the Siamese network to separate the target and background pixels. Therein a sparse and mixture strategy is used to construct rich training samples. Furthermore, Rao et al. [43] combined a transformer with the Siamese network to obtain good separation capacity, of which the background hybrid strategy is used to obtain class-balanced training sets. Besides, there are many other DL-based HTD algorithms, such as the generative adversarial networks (GAN)-based [44], [45], long short-term memory (LSTM)-based [46], and transformer-based detectors [47], which also achieve promising detection performances in many complex real scenes.

Although the effectiveness of the above DL-based HTD algorithms has been demonstrated, there are still some challenging problems worthy of further study.

1) Primarily, how to use HSI itself and limited prior targets to construct the balanced background and target training sets is a tremendous challenge for DL-based HTD.
2) Emphatically, how to highlight targets while suppressing backgrounds by the deep network is worth pondering.

To cope with these problems and challenges, in this article, a spectral aggregation and separation network (SASN) with a target band random mask (TBRM) is proposed to highlight targets and suppress backgrounds well. This work is mainly composed of three parts, i.e., training set preparation, training, and testing. In the training set preparation of SASN, we first develop a multifarious representative background selection strategy (MRBS) incorporating density-based spatial clustering of applications with noise (DBSCAN) [48] and SAM to obtain a background training set with negative label 0. Subsequently, to make the training sets balance, we propose a simple but effective data augmentation (DA) method, TBRM, to generate sufficient and reasonable target training set with positive label 1 by randomly repeating zero-masking the spectral bands of a prior target. In the training stage, the class-balanced training sets and a prior target are fed into the three-branch SASN based on residual connection and squeeze-and-excitation (SE) attention mechanism, which can automatically and fully extract the high discriminative features and nonlinear ones in the spectra. Besides, to strengthen the separation between the targets and the backgrounds, we present a triplet-soft loss function based on the Pearson correlation coefficient (PCC) in SASN. During testing, the trained SASN automatically distinguishes the similarities and differences between the to-be-detected pixels and the prior target, then its outputs construct the final detection result. It is worth emphasizing that the original HSI and only one prior target are required in SASN.

More plainly, the contributions of this article are summarized as follows.

1) A novel SASN with TBRM is proposed to tackle the problem of class imbalance and fulfill the target

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highlighting and background suppression simultaneously for HTD.

2) An MRBS is proposed to construct a multifarious and representative background set for SASN. For initial coarse classification, DBSCAN is first employed to obtain several clusters and a cluster of noises which is discarded. Subsequently, SAM is used for further selecting representative background pixels by calculating spectral angles.
3) For the target training set of SASN and the problem of class imbalance, the proposed simple but effective TBRM augments target samples by repeating randomly zero-masking spectral bands of a prior target. Notably, the samples augmented by TBRM can not only avoid mixing with manual interference information but also simulate spectral variability. Besides, they can force the network to extract global band information, thus increasing the generalization, stability, and overall performance of SASN.
4) To highlight targets and suppress backgrounds, the proposed SASN aggregates intraclass samples and separates interclass ones to distinguish spectral similarities and differences simultaneously. Moreover, a triplet-soft loss function based on PCC is presented in SASN to strengthen the separation of targets and backgrounds.

The remainder of this article is arranged as follows. In Section II, we briefly introduce the relevant knowledge involved in this article. We describe the principle of the proposed method in detail in Section III. Then, we verify the effectiveness and superiority of the proposed method through extensive experiments and analysis in Section IV. Finally, we summarize the work of this article in Section V.

II. Related Work

A. Density-Based Spatial Clustering of Applications With Noise

DBSCAN is a density-based unsupervised clustering, which divides data into a cluster of noises and other clusters with arbitrary shapes according to density relations without being affected by the spatial attribute. Therefore, DBSCAN is employed to discover the clusters and noises in HSI based on density relations determined by spectral similarity in this article.

There are two parameters in DBSCAN, i.e., the minimum radius (Eps) of the neighborhood of a pixel and the minimum number of contained points (MinPts) in the Eps-neighborhood. Suppose an HSI dataset is $\boldsymbol{X} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N] \in \mathbb{R}^{d\times N}$, where d and N are the numbers of spectral bands and pixels, respectively. The data space composed of $\boldsymbol{x}_k$ in $\boldsymbol{X}$ is represented as the set $\mathcal{X}$. The Eps-neighborhood of $\boldsymbol{x}_p$ in $\mathcal{X}$ is

$$\mathcal{D}_{\text{Eps}}(\boldsymbol{x}_p) = \{\boldsymbol{x}_q \in \mathcal{X} | \operatorname{dist}(\boldsymbol{x}_p, \boldsymbol{x}_q) \le \text{Eps}\} \tag{1}$$

where $\operatorname{dist}(\boldsymbol{x}_p, \boldsymbol{x}_q)$ is the distance between $\boldsymbol{x}_p$ and $\boldsymbol{x}_q$. The pixel amount contained in the d-dimensional hypersphere $\mathcal{D}_{\text{Eps}}(\boldsymbol{x}_p)$ is $N_{\text{Eps}}(\boldsymbol{x}_p)$. Besides, there are some definitions.

1) *Core Pixel:* $\boldsymbol{x}_p \in \mathcal{X}$ with $N_{\text{Eps}}(\boldsymbol{x}_p) \ge \text{MinPts}$.
2) *Border Pixel:* $\boldsymbol{x}_p \in \mathcal{D}_{\text{Eps}}(\boldsymbol{x}_q)$ with $N_{\text{Eps}}(\boldsymbol{x}_p) < \text{MinPts}$, where $\boldsymbol{x}_q$ is a core pixel.
3) *Noise Pixel:* The pixel that does not belong to the Eps-neighborhoods of any pixels.
4) *Directly Density-Reachable:* The pixel $\boldsymbol{x}_p \in \mathcal{D}_{\text{Eps}}(\boldsymbol{x}_q)$ is directly density-reachable from the core pixel $\boldsymbol{x}_q$.
5) *Density-Reachable:* Given a sequence of some pixels in $\mathcal{X}$, $\{\boldsymbol{y}_1, \boldsymbol{y}_2, \ldots, \boldsymbol{y}_k\}$, where $\boldsymbol{y}_1 = \boldsymbol{x}_q$ and $\boldsymbol{y}_k = \boldsymbol{x}_p$. For $\forall i = 1, \ldots, k-1$, if $\boldsymbol{y}_{i+1}$ is directly density-reachable from $\boldsymbol{y}_i$, $\boldsymbol{x}_p$ is called density-reachable from $\boldsymbol{x}_q$ (i.e., a transitive relation).
6) *Density-Connected:* $\boldsymbol{x}_p$ and $\boldsymbol{x}_q$ are density-connected if they are both density-reachable from another pixel $\boldsymbol{x}_o$.

At each iteration in DBSCAN, an unvisited core pixel $\boldsymbol{x}_{\text{core}}$ in $\mathcal{X}$ is randomly selected as the clustering center. Then, form a cluster by retrieving all unvisited pixels density-reachable from $\boldsymbol{x}_{\text{core}}$. After all core and border pixels are visited, the remaining pixels are determined as the cluster of noises.

B. Data Augmentation

Previous studies indicated that class imbalance has a significant detrimental effect on model convergence and generalization of the deep network [49], [50]. Since there are extremely imbalanced data, obtaining balanced training sets is the primary challenge of DL-based HTD. However, most common DA methods are not suitable for HSI in HTD. For instance, one of the simplest methods, under-sampling, randomly discarding the inherent samples in the majority class, could not be applied to HSI with rare positive samples. Besides, over-sampling, referring to creating new minority class samples to achieve class balance, also could not take effect [51]. In addition, since the input of pixel-wise-based HTD is the spectral vector, many general DA methods for ordinary visible images (e.g., RGB images), such as mirroring, rotation, folding, clipping, scaling, and shearing [52], [53], [54], may destroy the spectral structure and give rise to worse training effects.

Fortunately, there are some other DA methods, such as Cutout [55], Mixup [56], CutMix [57], and GridMask [58], that might be available to HSI since their occlusion ideas can be carried out on the spectral dimension of prior targets to augment the target training set. However, the way of mixing noises up with prior targets has little influence on the spectral signatures such that it cannot augment the richness of the training set, and it is easy to cause a waste of resources. The method of mixing backgrounds with prior targets requires manual adjustment of abundance, of which the setting has a great impact on the detection performances, and mixed information in the targets may result in the misjudgment of the detector. The means of replacing some bands of prior targets with those of backgrounds may produce some spectra that are not in line with the actual situation, which may harm the training results. Conversely, the variant Cutout without parameter adjustment may not make the spectra mixed with manual information. Accordingly, we propose a simple and effective TBRM to augment the target training set in SASN.

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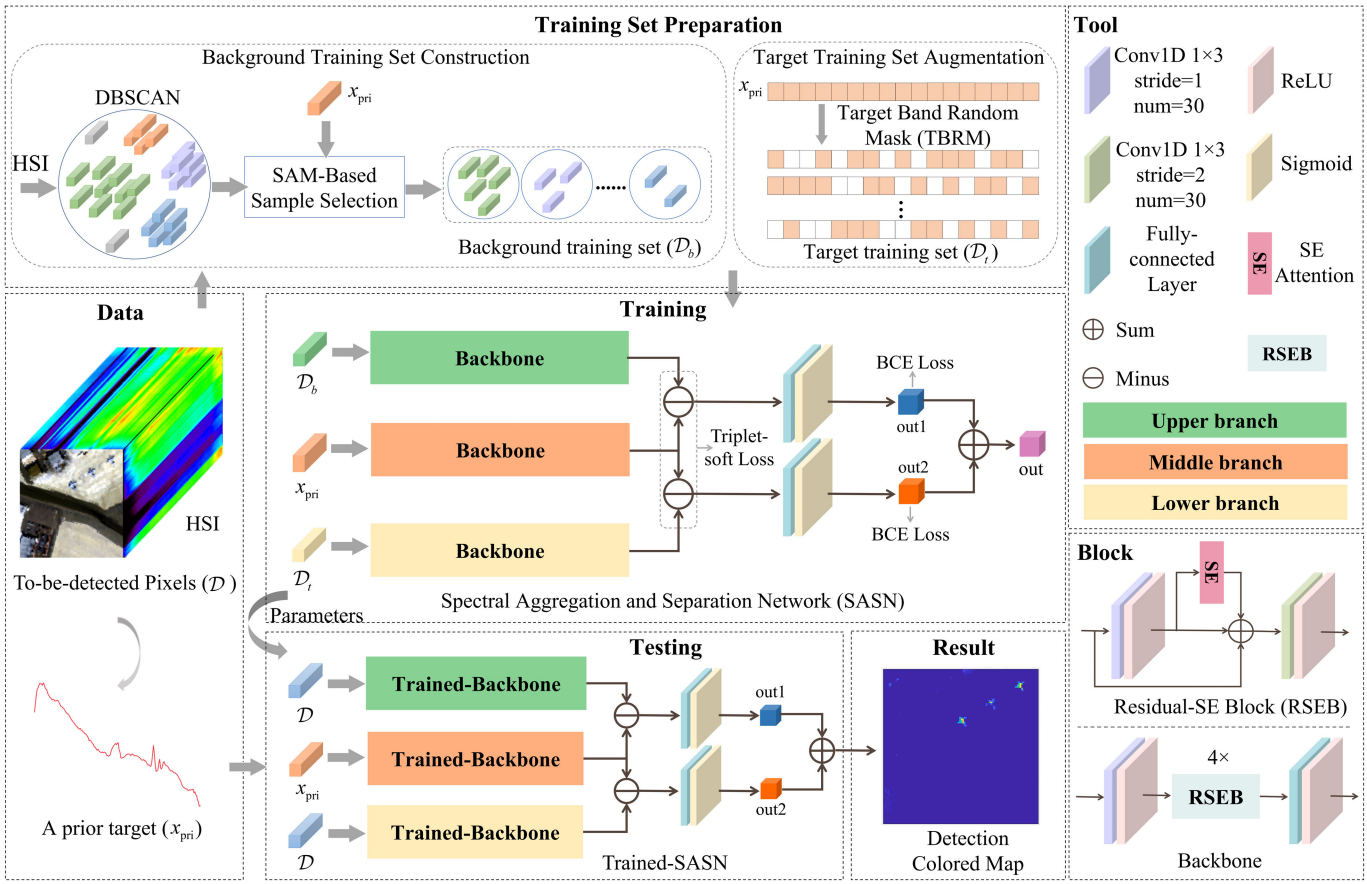


Fig. 1. Flowchart of the proposed method, including three parts: training set preparation (i.e., MRBS and TBRM), training of SASN, and testing of SASN.

III. Proposed Method

In this section, the proposed method is described in detail and the flow diagram is illustrated in Fig. 1, which is mainly divided into three parts, i.e., training set preparation, training, and testing. Considering the class imbalance in HSI, a balanced background training set (with negative label 0), and a target training set (with positive label 1) are constructed in the training set preparation by MRBS and TBRM, respectively. In the training stage, the training sets and the prior target are fed into the SASN, and the proposed triplet-soft loss function is used to separate the backgrounds and targets better. In the testing stage, the to-be-detected pixels and the prior target are fed into the trained SASN to obtain the final detection result. Notably, the backbone of the SASN is mainly composed of the residual-SE block (RSEB). The detailed implementation of the proposed method is described in Algorithm 1.

A. Background Training Set Construction

For the background training set, the MRBS consisting of two inseparable steps is proposed. It selects the pure and representative background training set from the multicategorical background pixels in HSI without multilabels.

Specifically, DBSCAN is first used to cluster the original HSI into n clusters and a cluster of noises $\mathcal{C}^{(0)}$. Since background pixels in HSI are much more than target pixels, the clustering centers in DBSCAN are generally background pixels with more neighbors, and they mainly cluster backgrounds. As Section I mentioned, target pixels are sparse with low densities, so some of them might be clustered as $\mathcal{C}^{(0)}$, which is discarded for preliminary purification.

Subsequently, in order to further select the representative background pixels with high purity, the simple but typical SAM is employed. In each iteration, SAM is used to calculate spectral angles between pixels in each cluster and a known pixel $\boldsymbol{x}_q$, herein, $\boldsymbol{x}_q = \boldsymbol{x}_{\text{pri}} \in \mathbb{R}^{d\times 1}$ is a given prior target in the beginning. Then, the pixels with maximum angles are selected as the background training samples in proportion. The specific procedure of MRBS is presented in Algorithm 2. Herein, the spectral angle $\theta(\boldsymbol{x}_p, \boldsymbol{x}_q)$ between the pixel $\boldsymbol{x}_p \in \mathbb{R}^{d\times 1}$ in HSI and $\boldsymbol{x}_q$ calculated by SAM is denoted as $\theta(\boldsymbol{x}_p, \boldsymbol{x}_q) = \arccos(\boldsymbol{x}_p^{\mathrm{T}}\boldsymbol{x}_q)/(\|\boldsymbol{x}_p\|\|\boldsymbol{x}_q\|)$, where $\|\cdot\|$ is l_2-norm, $(\cdot)^{\mathrm{T}}$ is the transpose operation. Notably, the larger the angle is, the more dissimilar the two pixels are. Besides, since samples in each cluster are unevenly distributed, the number of background pixels proportionally selected in $\mathcal{C}^{(i)}$ is $n_b^{(i)} = m^{(i)} \times (N_s/N)$, where $m^{(i)}$ and N_s are the number of pixels in $\mathcal{C}^{(i)}$ and final background/target training samples, respectively. Particularly, in order to select background samples from as many different classes as possible, $\boldsymbol{x}_q$ should be updated in time. After n iterations, the background training set $\mathcal{D}_b$ with negative label 0 is constructed for SASN, herein, the cardinality $|\mathcal{D}_b| = N_s$.

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Algorithm 1 SASN With TBRM for HTD

Input: The HSI and a prior target.
Output: The detection result of the proposed method.
1 **Background Training Set Construction:** Select pure and representative background pixels in the whole HSI via MRBS, and construct negative training set with label 0.
2 **Target Training Set Augmentation:** Adopt TBRM to augment sufficient target training samples, and construct positive training set with label 1 to class balance.
3 **Training:** Input the selected background training set, a prior target, and the generated target training set to the top, middle, and bottom branches of the SASN, respectively, and train SASN.
4 **Testing:** Feed the to-be-detected pixels into the top and bottom branches of the trained SASN identically, and a prior target is fed into the middle branch, then test and the outputs of the network construct the final detection result.

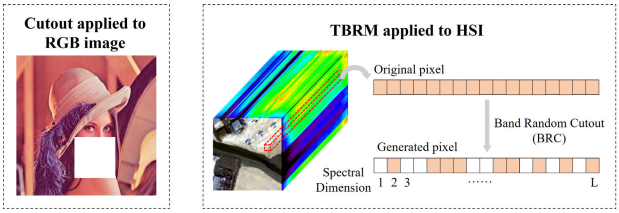


Fig. 2. Comparison between the original Cutout and the proposed TBRM.

In a word, the proposed MRBS selects as many classes of pure and representative background training samples as possible due to the combined two steps. First, some target pixels in HSI would be clustered into $\mathcal{C}^{(0)}$ by DBSCAN and discarded, to reduce the number of pixels used for the next selection and increase the purity of background pixels. Second, the SAM and the update strategy of $\boldsymbol{x}_q$ can not only make the selected background pixels the least similar to the prior target $\boldsymbol{x}_{\text{pri}}$ but also make the difference among themselves larger, resulting in the $\mathcal{D}_b$ being multifarious and representative.

B. Target Training Set Augmentation

The target pixels in HSI usually account for less than 1%, namely, the numbers of background and target training samples are seriously unbalanced. Therefore, it is vital to augment the target training set to deal with the class imbalance. Inspired by a general DA algorithm, Cutout [55], which is applied to ordinary visible images, the simple but effective TBRM is proposed to augment the target training set in this article. To the best of our knowledge, it is the first time to apply the variant Cutout to augment the samples in HSI.

As illustrated in Fig. 2, the original Cutout completely obstructs the entire square patch by zero-masking, while the TBRM augments samples in spectral dimension. To be specific, a new target training sample is obtained by zero-masking the corresponding values of some random spectral bands of a prior target $\boldsymbol{x}_{\text{pri}}$. After repeating the TBRM on $\boldsymbol{x}_{\text{pri}}$, ample target training samples are generated for class balance.

Concretely, for a given prior target $\boldsymbol{x}_{\text{pri}} = [l_1, l_2, \ldots, l_d]^{\text{T}}$, l_i $(i = 1, 2, \ldots, d)$ represents the value of the ith spectral band of $\boldsymbol{x}_{\text{pri}}$. Before generating a new sample, a set s $(0 \leq |s| \leq d)$ containing the serial numbers of the spectral bands to be zero-masked is randomly acquired. Then, the kth target sample $\boldsymbol{x}_{\text{pri}}^{(k)} = [l_1^{(k)}, l_2^{(k)}, \ldots, l_d^{(k)}]^{\text{T}}$ $(k = 1, 2, \ldots, N_s)$ is generated by zero-masking. Herein, the value $l_j^{(k)}$ $(j = 1, 2, \ldots, d)$ is determined as

$$l_j^{(k)} = \begin{cases} 0, & \text{if } j \in s \\ l_j, & \text{otherwise.} \end{cases} \tag{2}$$

Notably, there is a certain possibility that the values of continuous bands may be zero-masked. After N_s times of zero-masking of $\boldsymbol{x}_{\text{pri}}$, the target training set $\mathcal{D}_t$ with positive label 1 is finally constructed with $|\mathcal{D}_t| = N_s$ for SASN.

Significantly, in addition to being able to augment rich target training samples that are not mixed with manual interference information through simple random operations, TBRM has other advantages. First, each sample in $\mathcal{D}_t$ is randomly generated, which simulates the spectral variability in the real scene, thus increasing the ability to generalize the network. Moreover, although some important bands may be randomly zero-masked, the lost information may be recovered through the propagation of deep feature maps and other information in the deep network. It should be noted that this method forces the network to extract the features of global bands, instead of focusing on the local information, which increases the stability and overall performance of the network to a great extent.

Algorithm 2 MRBS

Input: $\boldsymbol{X}$-the input HSI, $\boldsymbol{x}_{\text{pri}} \in \mathbb{R}^{d\times 1}$-a given prior target, N_s-number of background (target) training samples.
Output: $\mathcal{D}_b$-the background training set.
1 Initialize target training set $\mathcal{D}_b = \varnothing$, a known pixel $\boldsymbol{x}_q = \boldsymbol{x}_{\text{pri}}$, a known sequence $\mathcal{Q} = \{\boldsymbol{x}_q\}$;
2 Obtain the clusters $\mathcal{C}^{(i)}$ by DBSCAN, $\mathcal{C}^{(i)} = \{\boldsymbol{x}_1^{(i)}, \boldsymbol{x}_2^{(i)}, \ldots, \boldsymbol{x}_{m^{(i)}}^{(i)}\}(i = 0, 1, \ldots, n)$;
3 Discard the cluster of noises $\mathcal{C}^{(0)}$;
4 **for** $i = 1 : n$ **do**
5 Calculate the spectral angles between pixels in $\mathcal{C}^{(i)}$ and $\boldsymbol{x}_q$ denoted as $\Theta^{(i)} = \{\theta(\boldsymbol{x}, \boldsymbol{x}_q) | \boldsymbol{x} \in \mathcal{C}^{(i)}\}$;
6 Select the pixels with the top $n_b^{(i)}$ maximum spectral angles in $\Theta^{(i)}$ as background training set $\mathcal{D}^{(i)} = \{\boldsymbol{x} | \boldsymbol{x} \in \mathcal{C}^{(i)}, \theta(\boldsymbol{x}, \boldsymbol{x}_q) \in \max_{n_b^{(i)}} \Theta^{(i)}\}$, where $|\mathcal{D}^{(i)}| = n_b^{(i)}$, and $\max_{n_b^{(i)}} \Theta^{(i)}$ refers to the set of the top $n_b^{(i)}$ maximum angles in $\Theta^{(i)}$;
7 $\mathcal{D}_b \leftarrow \mathcal{D}_b \cup \mathcal{D}^{(i)}$, $\mathcal{Q} \leftarrow \mathcal{Q} \cup \mathcal{D}^{(i)}$;
8 Average (AVG) the pixels in $\mathcal{Q}$ along the spectral dimension as the new known pixel $\boldsymbol{x}_q$.
9 **end**

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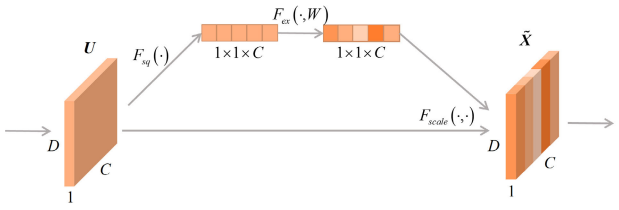


Fig. 3. SE block.

C. Spectral Aggregation and Separation Network

Aiming at highlighting the targets and suppressing the backgrounds, the SASN is proposed in this article, which includes the top, middle, and bottom branches with consistent structure and parameter settings. In particular, the input of each branch in SASN is a spectral vector, and a prior target is fed into the middle branch during both training and testing.

In SASN, each branch contains nine 1 × 3 1-D convolutions, each of which is followed by the rectified linear unit (ReLU) layers (denoted as Conv-ReLU). These 1-D convolutions are with kernel number of 30 and a stride of 1 or 2. Notably, the convolutions with stride 2 are used for feature compression, and the ReLU layers are used to increase the nonlinear expression ability of the network. Specifically, the SE attention module, two Conv-ReLU (with strides 1 and 2), and the residual connection construct the RSEB. The backbone of the SASN is composed of a Conv-ReLU with stride 1, four successive RSEB, and fully connected (FC) layers followed by ReLU, to thoroughly tap high discriminative spectral features and nonlinear ones. Particularly, the SE attention module is shown in Fig. 3, including a global AVG pooling layer and two FC layers. Through "Squeeze $F_{\mathrm{sq}}(\cdot)$–Excitation $F_{\mathrm{ex}}(\cdot, W)$–Channel-Wise Multiplication $F_{\mathrm{scale}}(\cdot,\cdot)$," the weight of each channel in the feature map is different, which contributes to pay more attention to the feature channels with high weights in SASN.

To effectively distinguish the similarities and differences between spectra, the outputs of the backbone of the three branches are operated as

$$\begin{aligned} \boldsymbol{f}_{\text{difference}} &= \boldsymbol{f}^{p} - \boldsymbol{f}^{b} \\ \boldsymbol{f}_{\text{similarity}} &= \boldsymbol{f}^{p} - \boldsymbol{f}^{t} \end{aligned} \tag{3}$$

where $\boldsymbol{f}^{b}$, $\boldsymbol{f}^{p}$, and $\boldsymbol{f}^{t}$ are the outputs of the top, middle, and bottom branches, respectively. Then the feature maps representing the spectral differences and similarities are obtained, which are denoted as $\boldsymbol{f}_{\text{difference}}$ and $\boldsymbol{f}_{\text{similarity}}$, respectively. Subsequently, the other FC layer followed by the sigmoid function is used to classify the background and target pixels, the outputs of which are denoted as $\boldsymbol{f}'_{\text{difference}}$ and $\boldsymbol{f}'_{\text{similarity}}$, respectively.

Finally, the output of the SASN is obtained by

$$\boldsymbol{f}_{\text{output}} = \boldsymbol{f}'_{\text{difference}} + \boldsymbol{f}'_{\text{similarity}}. \tag{4}$$

Inspired by the Soft-PN loss function [59], the triplet-soft loss function based on PCC is presented in HTD for the first time in this article. The triplet-soft loss makes the negative samples not only farther from the prior target but also the positive samples in the feature space, to strengthen the spectral intraclass aggregation and interclass separation.

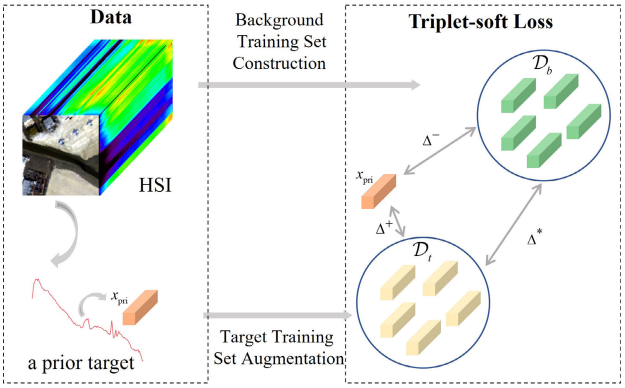


Fig. 4. Illustration of the proposed triplet-soft loss.

It is worth noting that since some bands of target training sets generated by TBRM are zero, the corresponding values of some dimensions of the feature maps after the SASN are maybe zero, too. To reduce the influence of some special values when calculating the similarity coefficient, the PCC, which is insensitive to absolute numbers, is adopted in the triplet-soft loss function. The PCC is described as

$$\rho = \frac{\boldsymbol{u}^{\mathrm{T}}\boldsymbol{v}}{\left\|\boldsymbol{u} - \bar{u}\mathbf{1}\right\|\left\|\boldsymbol{v} - \bar{v}\mathbf{1}\right\|} \in [-1, 1] \tag{5}$$

where $\boldsymbol{u} = [u_1, u_2, \ldots, u_M]^{\mathrm{T}}$, $\boldsymbol{v} = [v_1, v_2, \ldots, v_M]^{\mathrm{T}}$, $\bar{u} = (1/M)\sum_{i=1}^{M} u_i$, $\bar{v} = (1/M)\sum_{i=1}^{M} v_i$, and $\mathbf{1}$ denotes the all-ones column vector with a matching dimension. It is noteworthy that if the absolute value of ρ, denoted as $|\rho|$, is closer to 1, the similarity between the two vectors is higher, otherwise, bottom.

Based on the PCC mentioned above, the triplet-soft loss function is expressed as

$$\mathrm{Loss}_{\mathrm{TS}} = \frac{1}{B}\sum_{i=1}^{B}\left[\left(\frac{e^{\Delta_i^+}}{e^{\Delta_i^{\max}} + e^{\Delta_i^+}} - 1\right)^2 + \left(\frac{e^{\Delta_i^{\max}}}{e^{\Delta_i^{\max}} + e^{\Delta_i^+}}\right)^2\right] \tag{6}$$

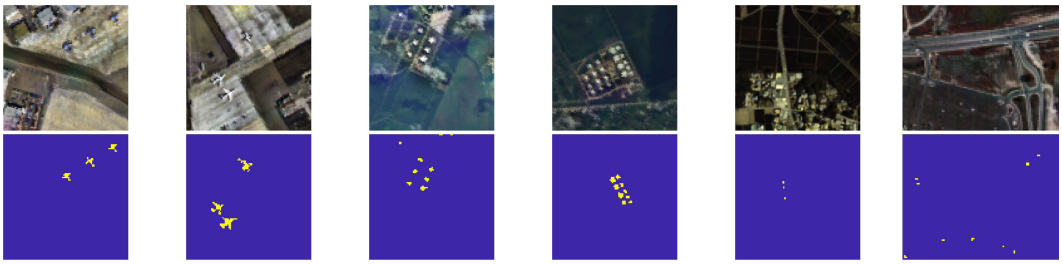
where $\Delta_i^+ = |\rho(\boldsymbol{f}_i^p, \boldsymbol{f}_i^t)|$, $\Delta_i^- = |\rho(\boldsymbol{f}_i^p, \boldsymbol{f}_i^b)|$, $\Delta_i^* = |\rho(\boldsymbol{f}_i^t, \boldsymbol{f}_i^b)|$, $\Delta_i^{\max} = \max(\Delta_i^-, \Delta_i^*)$, and B is batch size. Besides, $\boldsymbol{f}_i^b$, $\boldsymbol{f}_i^p$, and $\boldsymbol{f}_i^t$ represent the ith feature vector outputs of the top, middle, and bottom branches after the backbone in one epoch, respectively.

As illustrated in Fig. 4, the triplet-soft loss can make the distance in the feature space between the positive samples and the prior target smaller, the distance between the negative samples both and the prior target and positive samples larger, namely, Δ^+ is as small as possible, and Δ^- and Δ^* are as larger as possible.

Furthermore, since the HTD is considered as the binary classification, the binary cross-entropy (BCE) loss function is necessarily used in SASN to classify the targets and backgrounds, which is

$$\mathrm{Loss}_{\mathrm{BCE}} = -\frac{1}{B}\sum_{i=1}^{B}\left[y_i \cdot \log f_i + (1 - y_i) \cdot \log(1 - f_i)\right] \tag{7}$$

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(a) (b) (c) (d) (e) (f)

Fig. 5. Pseudo-color images and ground-truth maps of six datasets. (a) San Diego-1. (b) San Diego-2. (c) Urban-1. (d) Urban-2. (e) Chikusei. (f) HYDICE.

where f_i is the output of the sigmoid function, and y_i refers to the positive or negative labels 1 or 0.

Finally, the total loss in SASN is the sum of the triplet-soft loss with weighted coefficient β and BCE loss, which is given as $\text{Loss} = \beta \text{Loss}_{\text{TS}} + \text{Loss}_{\text{BCE}}$, where the β would be adjusted according to the actual application.

D. Under-Text Pixels' Testing

After obtaining the trained SASN, the to-be-detected pixels $\boldsymbol{x}_k \in \mathbb{R}^{d\times 1}$ $(k = 1, 2, \ldots, N)$ in HSI are fed into it to detect. The structure of SASN during the testing stage is consistent with that during the training stage. In the testing phase, $\boldsymbol{x}_k$ is fed into the top and bottom branches similarly, and the prior target $\boldsymbol{x}_{\text{pri}}$ is input into the middle branch. Then, the SASN automatically distinguishes the spectral similarity and difference between $\boldsymbol{x}_k$ and $\boldsymbol{x}_{\text{pri}}$.

The output of SASN, i.e., the detection fraction of $\boldsymbol{x}_k$ is denoted as $D_{\text{SASN}}(\boldsymbol{x}_k)$, which represents the probability that $\boldsymbol{x}_k$ is the target pixel. All the $D_{\text{SASN}}(\boldsymbol{x}_k)$ constitute the final detection result of SASN. Finally, $\boldsymbol{x}_k$ is determined as a target or not by setting dynamic threshold η. If $D_{\text{SASN}}(\boldsymbol{x}_k) > \eta$, $\boldsymbol{x}_k$ is determined to be the target, otherwise, the background.

IV. Experimental Results and Analysis

Extensive experiments are carried out on six public HSI datasets to confirm the effectiveness and superiority of the proposed method. Sections IV-A and IV-B present the particulars of six datasets and evaluation indicators used in our experiments. Section IV-C introduces the experimental details and the optimal parameter settings. Section IV-D analyzes the parameter settings. Ablation experiments are conducted in Section IV-E to verify the feasibility of the corresponding module in SASN. In Section IV-F, the detection results and analysis of ten comparison detectors and SASN on six datasets are given to prove the superiority of the proposed method. Section IV-G gives the analysis of experimental time cost. The hardware environment in our experiments is Rayon R7-5800H Octa-core CPU, 3.20 GHz and 16 GB of random access memory, NVIDIA GEFORCE RTX 3060 GPU, and the operating system is Windows 11. DL-based experiments are implemented based on Python 3.9.12, Conda 4.12.0, and PyTorch 1.2.0. Other experiments are executed based on MATLAB R2019a.

A. Experimental Hyperspectral Datasets

To evaluate the effectiveness of the proposed method, six public HSI datasets are used in our experiments. Their pseudo-color images and ground-truth maps are shown in Fig. 5.

1) San Diego Datasets: The first two datasets, San Diego-1 and San Diego-2, are the subimages from the public HSI dataset collected by airborne visible/infrared imaging spectrometer (AVIRIS) [60] at San Diego Airport, CA, USA, on 9 November 2011. The spatial resolution is 3.5 m, and 189 spectral bands are reserved after removing low-quality bands, such as water vapor absorption bands and low signal-to-noise ratio (SNR) bands [61]. These two datasets are composed of 100 × 100 pixels, containing 64 and 134 pixels of airplanes as targets of interest, respectively.

2) Urban Datasets: The third and fourth datasets, Urban-1 and Urban-2, are from the Airport–Beach–Urban (ABU) dataset [62]. The corresponding scenes are the coastal area of Texas, USA, acquired by AVIRIS on 29 August 2010. Their spatial resolution is 17.2 m. After removing the low-quality spectral bands, there are 204 and 207 spectral bands contained in Urban-1 and Urban-2, respectively. Both datasets consist of 100 × 100 pixels, including nine buildings composed of 67 and 88 pixels as targets of interest, respectively.

3) Chikusei Dataset: The fifth dataset, Chikusei [63], was captured by Headwall HyperSpec-Vnir-C on 29 July 2014, over the junction between rural and urban areas in Tsukishi, Ibaraki, Japan. The spatial resolution is 2.5 m, and the 128 spectral bands are retained to increase SNR after leaving out the low-quality bands. The dataset contains 150 × 150 pixels, of which 15 pixels constitute the targets of interest located on the highway in the scene.

4) HYDICE Dataset: The sixth dataset, Hyperspectral Digital Imagery Collection Experiment (HYDICE), located in the urban area of California, USA, comes from the hyperspectral dataset obtained from the HYDICE sensor. The test dataset used in this article is a subimage containing 80 × 100 pixels, with a spatial resolution of 2 m. After removing the low-quality bands affected by water vapor and atmosphere, 162 spectral bands remained in our experiments. There are 21 pixels composed of cars and roofs regarded as targets of interest.

B. Evaluation Indicators

Three evaluation indicators are used in our experiments to quantitatively demonstrate the effectiveness and advantages of the proposed method.

1) Receiver Operating Characteristic: The receiver operating characteristic (ROC) curve is generally used to test the

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validity of HTD algorithms. Different abscissas and ordinates of ROC are obtained through dynamic threshold settings. The abscissa represents the probability that negative samples are incorrectly detected as positive samples in the detection, i.e., false alarm rate P_f. The vertical ordinates represent the probability that positive samples are successfully detected, i.e., true positive rate P_d. The well-performing HTD with high P_d and low P_f reflects the ROC curve more closing to the top left.

2) Area Under the ROC Curve: Since the ROC curves of various detectors are easy to overlap, the performances cannot be properly judged, and the area under the ROC curve (AUC) value is used complementally to quantitatively evaluate the detection performances [64]. AUC is obtained by integrating the ROC curve along the abscissa, that is, it represents the area between the ROC and the abscissa axis, and its value is between 0 and 1. Therefore, when the ROC curve is nearer to the top left corner, the AUC value is larger, and the performance is better. Combining ROC and AUC can assess the performances of the HTD algorithms more reasonably.

3) Separability Map: The separability map [65] is composed of a green box and an orange box, which, respectively, represent the range of normalized statistical values of background and target from the detection results. The top and bottom of the main box are composed of 25% and 75% of the normalized statistical value. The center mark of the box represents the median, that is, the 50th percentile. The whiskers on both sides of the box must be extended to 0.5% and 99.5% of the statistical value range, respectively [66]. The separability map is used to represent the separation degree of target and background detection values. When the two main boxes overlap, the values of targets and backgrounds are mixed badly; when the larger the gap between boxes, the better their separation is. Therefore, the separability map can be used to assess the degree of separation between the backgrounds and the targets in the detection results of HTD, and it can also be used to observe the distribution of data, which can verify the degree of background suppression [67].

C. Experiment Details and Settings

Note that all the detection results are the AVG values obtained from experiments ten times to ensure fairness.

1) Comparison Algorithms and Parameter Settings: To further explore the competitiveness of the proposed method, ten state-of-the-art comparison HTD algorithms are conducted in our experiments, including the following: three classic detectors, i.e., SAM [11], ACE [14], and CEM [22]; four ML-based detectors, i.e., SRTD [33], CSCR [34], SSBSTD [35], and TSETD [36]; a DL- and ML-based detector, self-spectral learning with GAN-based spectral–spatial detector (SSTD) [44]; and two DL-based detectors, i.e., HTD-Net [41] and TSCNTD [42]. For fairness, the settings of all the comparison detectors agree with the original work, and the parameters bound up with the platform are adjusted to the best by cross-validation.

Due to the different sizes and numbers of targets in various datasets, the inner and outer window sizes of the dual concentric window used in SRTD and CSCR are also inconsistent. The inner and outer window size settings for SRTD and CSCR detectors in our experiments are as follows: 11×11 and 21×21 for San Diego-1, 17×17 and 19×19 for San Diego-2, 7×7 and 9×9 for Urban-1 and Chikusei, and 5×5 and 7×7 for Urban-2 and HYDICE.

TABLE I
OPTIMAL PARAMETER SETTINGS FOR DBSCAN ON SIX DATASETS

PARAMETERS	Eps	MinPts
San Diego-1	0.30	12
San Diego-2	0.10	16
Urban-1	0.35	8
Urban-2	0.35	8
Chikusei	0.15	22
HYDICE	0.10	18

2) Parameter Settings of the Proposed Method: In the proposed method, the prior target of each dataset that remains constant is obtained by randomly taking five target pixels from each dataset and averaging them in the spectral dimension. The hyperparameters of SASN, i.e., the batch size, learning rate, and epoch are set to 256, 10^{-5}, and 400, respectively. For analyzing the loss function, we increment the β from 0.1 to 10, and when the β is equal to one, the detection performance is the best. In addition, there are three key parameters, the number of background/target training samples N_s, Eps, and MinPts, whose specific experiments and analysis are discussed in Section IV-D. According to the experimental results, the optimal setting of N_s is 1000, and the Eps and MinPts settings of different datasets are shown in Table I.

D. Parameter Analysis

1) Analysis of Parameters in DBSCAN: According to DBSCAN, when Eps is set too small, namely, the number of pixels contained in the neighborhood is small, most pixels are easy to be recognized as noise pixels. When Eps or MinPts is set too large, it is easy to recognize most pixels as separate clusters since there are many pixels in the neighborhood. When MinPts is set to small, most pixels will be clustered into the same cluster. In this section, we analyze the optimal settings and sensitivity of the two key parameters, Eps and MinPts in DBSCAN. The colored surfaces in Fig. 6 illustrate the AUC values of the proposed SASN under different DBSCAN parameter settings on six HSI datasets.

As shown, different settings of parameters have low sensitivity on the AUC values of San Diego-1, Urban-1, and Urban-2 in SASN, but there are still peaks in a certain range. For San Diego-1, when MinPts and Eps are set to 12 and 0.3, respectively, the AUC value is up to 0.99899. For Urban-1 and Urban-2, when the MinPts and Eps are set to 8 and 0.35, respectively, the AUC values are 0.99381 and 0.99948. Among them, when the MinPts and Eps are set to 14 and 0.35 on Urban-2, the AUC value also reaches 0.99948, but its clustering time is longer. For San Diego-2, Chikusei, and HYDICE, when Eps is large, the AUC value is low, namely, the sensitivity to parameter settings is relatively poor. When MinPts and Eps are set to 16 and 0.35 for San Diego-2, the AUC value reaches a local maximum, of 0.99785. For

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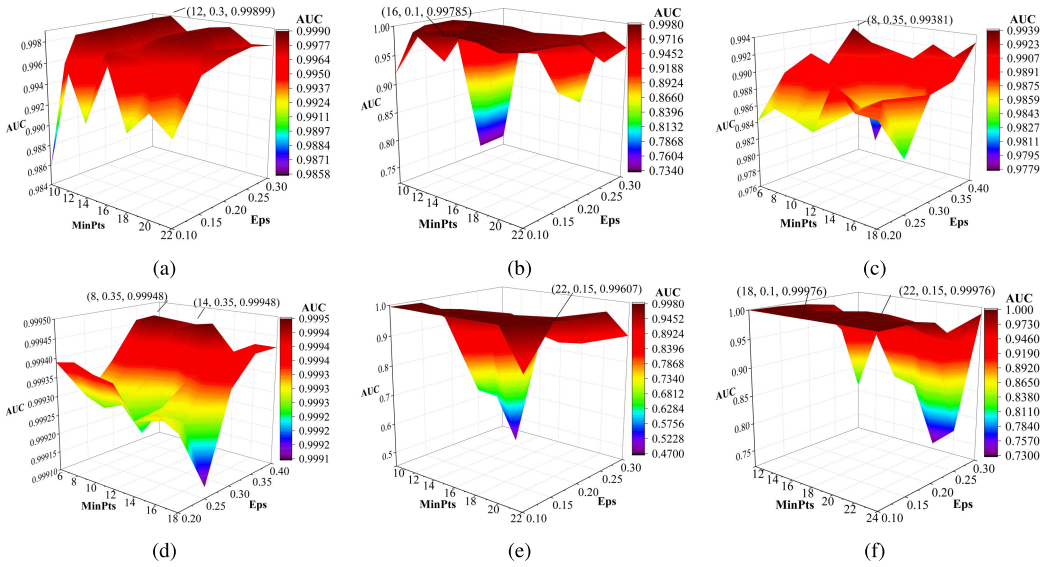


Fig. 6. Colored surfaces in terms of AUC values under different settings of the MinPts and Eps on six benchmark datasets. (a) San Diego-1. (b) San Diego-2. (c) Urban-1. (d) Urban-2. (e) Chikusei. (f) HYDICE.

TABLE II

AUC VALUES IN TERMS OF ABLATION EXPERIMENTS OF THE PROPOSED SASN WITH ITS VARIANTS ON SIX BENCHMARK DATASETS

AUC	San Diego-1	San Diego-2	Urban-1	Urban-2	Chikusei	HYDICE
SAM-Ablation[1]	0.99498	0.99291	0.98972	0.99823	0.99226	0.99604
CUTMIX[2]	0.99682	0.97771	0.98485	0.99945	**0.99707**	0.99946
MIXTURE[2]	0.86311	0.70265	0.65215	0.99826	0.99649	0.70978
BCELOSS[3]	0.99259	0.96711	0.98844	0.99934	0.98281	0.99428
Soft-PN[3]	0.99722	0.99058	0.99024	0.99910	0.98423	0.99771
SE-Ablation[4]	0.99768	0.99668	0.98488	0.99941	0.96886	0.99935
SASN	**0.99899**	**0.99785**	**0.99381**	**0.99948**	0.99607	**0.99976**

[1] SAM-Ablation is the ablation of the SAM used in MRBS.
[2] Cutmix and Mixture are the ablations of target training set augmentation.
[3] BCELoss and Soft-PN are the ablations of the loss function.
[4] SE-Ablation is the ablation of the network model.

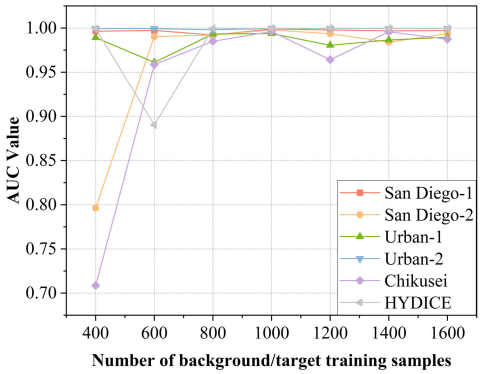


Fig. 7. AUC values of the proposed SASN under different background/target training sample number settings.

Chikusei and HYDICE, the AUC values reach the best, 0.99607 and 0.99976, when MinPts and Eps are set to 22 and 0.15, respectively. Although for HYDICE, the local optimum of AUC value can also be achieved when MinPts and Eps are 18 and 0.10, respectively, it takes a longer time.

2) Analysis of Training Sample Number: The number of background/target training samples N_s is another key parameter in the proposed method. Fig. 7 illustrates the AUC values of the SASN with different N_s settings on different datasets. It is obvious that for San Diego-2, Chikusei, and HYDICE datasets, when N_s is too small, the corresponding AUC values are also low, that is because the small number of training samples cannot make the network fully train. For other datasets, the AUC values are less sensitive to the setting of N_s, whereas, when $N_s = 1000$, SASN has the best detection results for all datasets on the whole.

Compared with other DL-based HTD algorithms, according to the original work, CNNTD and HTD-Net require a huge amount of training samples. Although the corresponding N_s of TSCNTD is also set to 1000 for a prior target, the original algorithm utilizes five prior targets, and the number of training samples used is relatively large. In contrast, the amount of training samples used in the proposed SASN is much reduced.

E. Ablation Study

In this section, we set six variants to confirm the feasibility of different components in the proposed method. For fairness, except for the part of the discussion module, the other structure and parameter settings are consistent. The ROC curves, AUC values, and separability map in terms of ablation experiments of the proposed SASN with its variants on six datasets are shown in Fig. 8, Table II, and Fig. 9, respectively.

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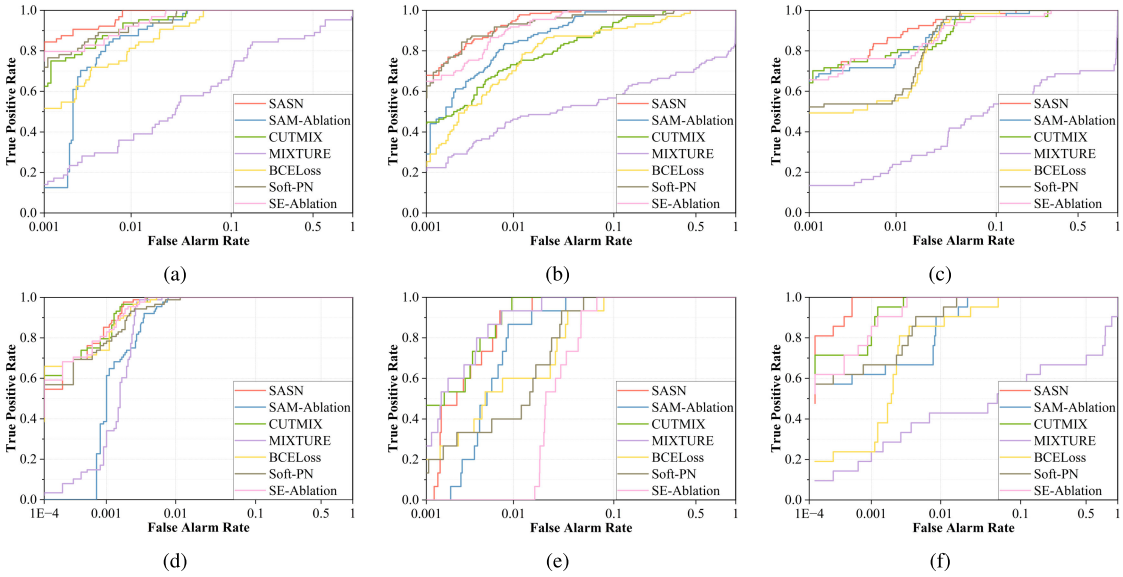


Fig. 8. ROC curves in terms of ablation experiments of the proposed SASN with its variants on six benchmark datasets. (a) San Diego-1. (b) San Diego-2. (c) Urban-1. (d) Urban-2. (e) Chikusei. (f) HYDICE.

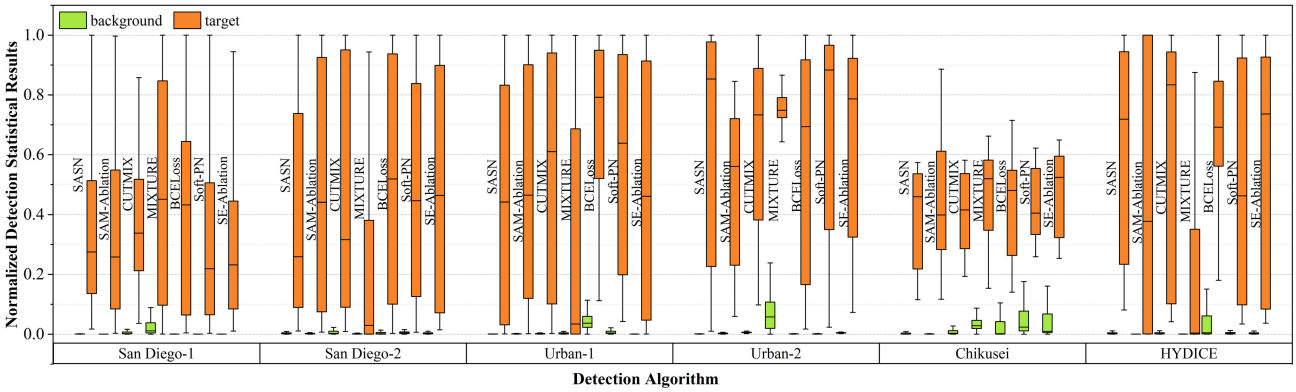


Fig. 9. Separability map in terms of ablation experiments of the proposed SASN with its variants on six benchmark datasets.

1) Ablation Analysis of SAM Used in MRBS: We conduct an ablation experiment on SAM used in MRBS. The ablation model is named SAM-Ablation, which removes SAM and uses the method of randomly selecting background pixels in each cluster proportionally instead. Although the differences between SAM-Ablation and SASN cannot be intuitively observed from the separability map, it can be clearly seen from the AUC values and ROC curves that the detection results of SAM-Ablation are always slightly inferior. This is because random selection always has uncontrollable problems, such as selecting similar pixels or repeatedly selecting the same pixel. The proposed MRBS utilizes superior SAM and prior target for measurement, selecting higher-quality background training samples to furtherance subsequent detection. However, SAM-Ablation also achieves relatively good performance on the whole, which further demonstrates the overall superiority of the proposed method from another perspective, that is, the mild decrease in background sample quality would not have a significant impact on the overall performance. In addition, we analyze the performance of SAM in Section IV-F to further validate the effectiveness and superiority of SAM.

2) Analysis of Target Training Set Augmentation: We compare and analyze the detection results corresponding to the other two DA methods. The first one is called CutMix, which generates new target training samples by replacing the spectral values of the random band of a prior target with those of the corresponding band of the selected background training samples. The second is the random hybrid method proposed in TSCNTD, that is, adding the selected background training samples with 0%–20% random abundance to the prior targets to construct new target training samples, which is called MIXTURE in our experiments.

As the ROC curves show, except for Chikusei, the ROC curve corresponding to the SASN is the closest to the top left, and the true positive rate first reaches 1. However, for the Chikusei dataset, the ROC curves corresponding to the three methods overlap to a large degree and have the same variation trend. Although the ROC curve of the SASN is not optimal, as the AUC values are shown, it has a narrow margin from the optimal value. In addition, from the separability map, SASN has the best background suppression for all datasets, while MIXTURE causes the worst background suppression. Among the experimental results of six datasets, the SASN achieves the best detection performances on the whole, due to the target training samples generated are pure and not mixed with manual interference information. The second is

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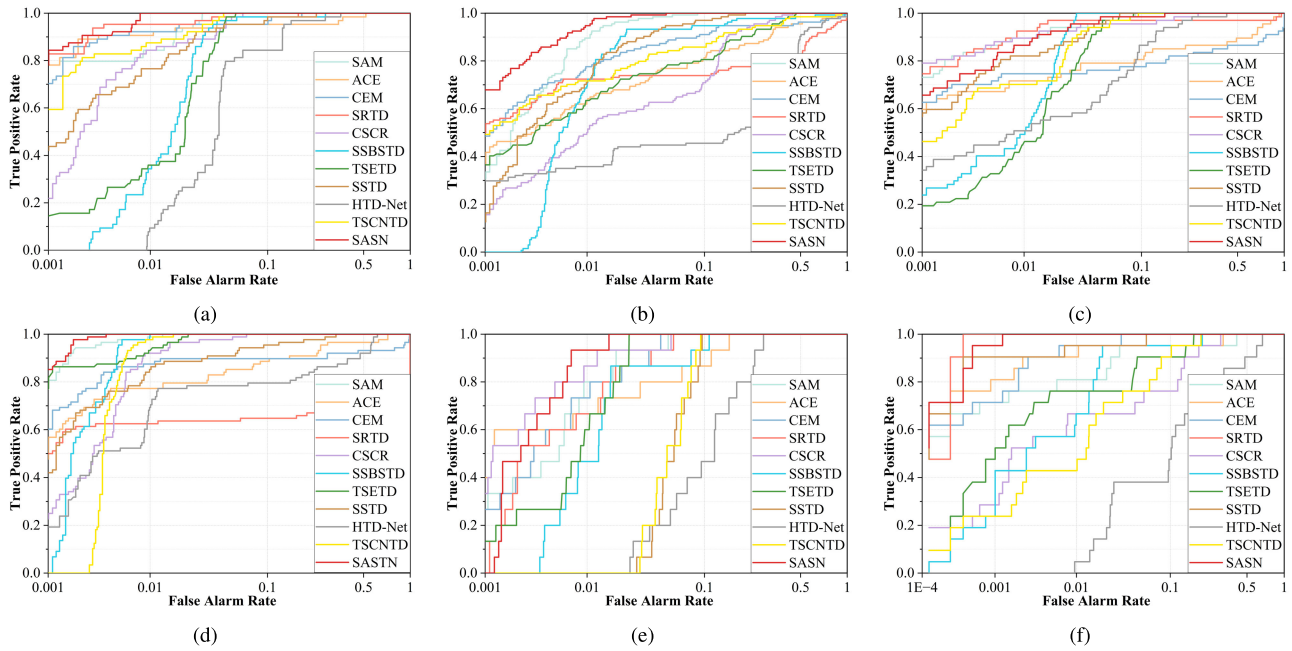


Fig. 10. ROC curves in terms of comparison experiments of ten different detectors on six benchmark datasets. (a) San Diego-1. (b) San Diego-2. (c) Urban-1. (d) Urban-2. (e) Chikusei. (f) HYDICE.

CutMix, the target training samples obtained by it only mix the background information in some random bands, so the detection performance is less affected by the background information. On the whole, the MIXTURE performs worst, since each band of the target training samples is mixed with the background spectral information, which makes the target training samples greatly affected by the background information.

3) Analysis of Loss Function: In order to verify the validity of triplet-soft loss, we compare the detection results of the SASN with those of the one without triplet-soft loss and those of the one with the original Soft-PN and BCE loss. As shown, the red curve corresponding to the SASN which utilizes both loss functions is closer to the top left. Although the ROC curves of some datasets overlap, the red curve performs better on the whole. Moreover, the curves using the original Soft-PN loss perform overall better than those using BCE loss alone. As is shown in the separability map, the gaps between the orange and green boxes of the three methods are far away, that is, the targets and the backgrounds are well separated, but the background suppression of the SASN is better. Yet for most datasets, the background suppression effect of BCE loss alone is generally poor.

It can be seen from the quantitative analysis that the detection performances of using two loss functions are generally better than that of using only BCE loss. This is because the triplet-soft loss can force the network to aggregate the target training set and prior target, while the background training set is separated both from the prior target and target training set, thus enhancing the separability. In addition, since the generated target training samples contain a large number of bands with values of 0, the PCC in the proposed triplet-soft loss is more applicable. Therefore, the detection performances of using the proposed triplet-soft loss are better than those of using the original Soft-PN loss. The above analysis proves the effectiveness of the triplet-soft loss proposed in SASN, which makes it more capable of suppressing background pixels and achieving better detection results.

4) Ablation Experiment of Network: Ablation experiments are carried out on the SE attention mechanism module in SASN. Except for the partial overlap on Urban-1 and Urban-2, the red curves corresponding to the SASN are located on the top part of the blue curves on balance, and their true positive rates reach 1 earlier. From the separability map, in most cases, whether the SE attention is included or not both can achieve better separation and background suppression. However, the SASN with SE attention module performs better on the whole. That is because compared with the Conv-ReLU, the SE attention module pays more attention to the most informative part of the channel dimension, and the two complement each other, making the network more fully extract the key information to increase the performance of the SASN. Therefore, the effectiveness of the SE attention mechanism in SASN is proved.

F. Analysis and Comparison of the Detection Results

We analyze and compare the experimental detection results of ten comparison algorithms and the proposed method from both quantitative and visual evaluation.

1) Quantitative Analysis: Fig. 10, Table III, and Fig. 11 exhibit the ROC curves, AUC values, and separability maps according to the quantitative detection results of ten various algorithms on the six baseline datasets, respectively.

Say concretely, the first detector SAM, which is used in our proposed method, is superior to the other two traditional detectors in terms of AUC values and ROC curves but performs poorly in the HYDICE dataset. From the separability maps, it can be seen that SAM performs extremely well in target highlighting, which shows that its ability to match and

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TABLE III
AUC VALUES IN TERMS OF COMPARISON EXPERIMENTS OF TEN DIFFERENT DETECTORS ON SIX BENCHMARK DATASETS

AUC	San Diego-1	San Diego-2	Urban-1	Urban-2	Chikusei	HYDICE
SAM	0.99623	0.99519	**0.99471**	0.99899	0.99132	0.97649
ACE	0.99317	0.93792	0.98986	0.95330	0.98377	0.98429
CEM	0.99693	0.96683	0.96915	0.92359	0.99307	0.99765
SRTD	0.99568	0.84781	0.97077	0.68563	0.98901	0.99973
CSCR	0.99206	0.93899	0.98899	0.99470	0.99361	0.95269
SSBSTD	0.97958	0.97159	0.98943	0.99726	0.97911	0.98402
TSETD	0.98093	0.94639	0.98272	0.99787	0.98988	0.97813
SSTD	0.98778	0.98708	0.99271	0.98403	0.94099	0.99680
HTD-Net	0.94726	0.75527	0.95344	0.91348	0.87963	0.83482
TSCNTD	0.99547	0.95357	0.98916	0.99570	0.94408	0.96916
SASN	**0.99899**	**0.99785**	0.99381	**0.99948**	**0.99607**	**0.99976**

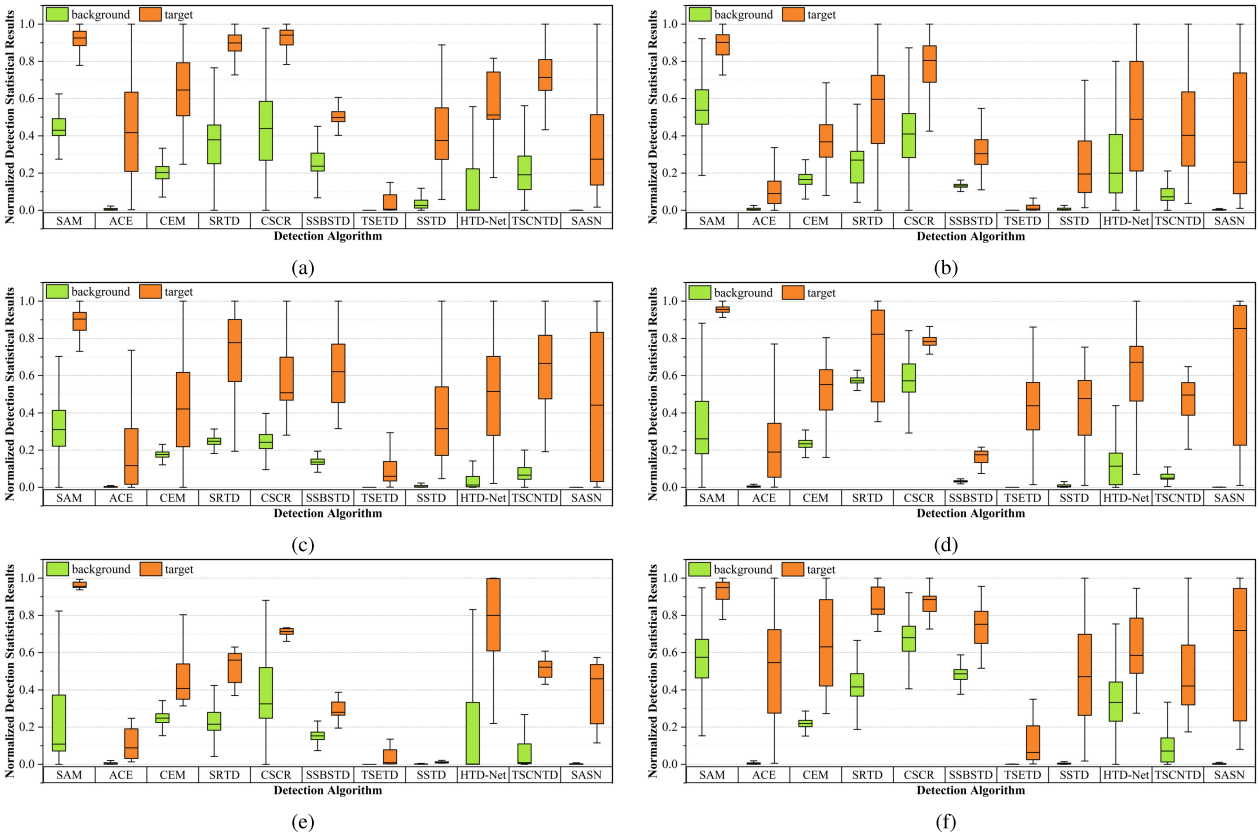


Fig. 11. Separability maps in terms of comparison experiments of ten different detectors on six benchmark datasets. (a) San Diego-1. (b) San Diego-2. (c) Urban-1. (d) Urban-2. (e) Chikusei. (f) HYDICE.

detect by calculating spectral angles is strong. Moreover, the trends of ROC curves corresponding to the other two classic detectors ACE and CEM are relatively consistent, and their AUC values are general and erratic. On San Diego-2 and two Urban datasets, they perform poorly since there are more subpixels in these datasets. More particularly, the green box of ACE is closer to both the ordinate 0 and the orange one for most datasets, which means that it has excellent background suppression capacity but indifferent separation ability. Furthermore, the four ML-based detectors exhibit unstable AUC values since they rely on expert knowledge and manual participation, among which SRTD is the most volatile. The two boxes of SRTD and CSCR are both at a higher level than most other algorithms, suggesting that their target-highlighting ability is extraordinary, but their background suppression is weak. This is because the local background dictionaries obtained by the dual concentric window in SRTD and CSCR are easily polluted by the target pixels, leading to a high false alarm rate. For most datasets, the target and background detection results of SSBSTD are in the middle level, and target highlighting and background suppression are very ordinary. In particular, it has a stable separation ability, since the target and background sparse vectors are calculated separately. It can be clearly seen from the separability maps that TSETD has a very superior background suppression ability, but its orange box is basically at the lowest level, which proves that its target detection and highlighting ability is the weakest. Obviously, these ML-based detectors could not achieve the best background suppression and target-highlighting capabilities at the same time. In the SSTD and the two DL-based detectors, the AUC values level

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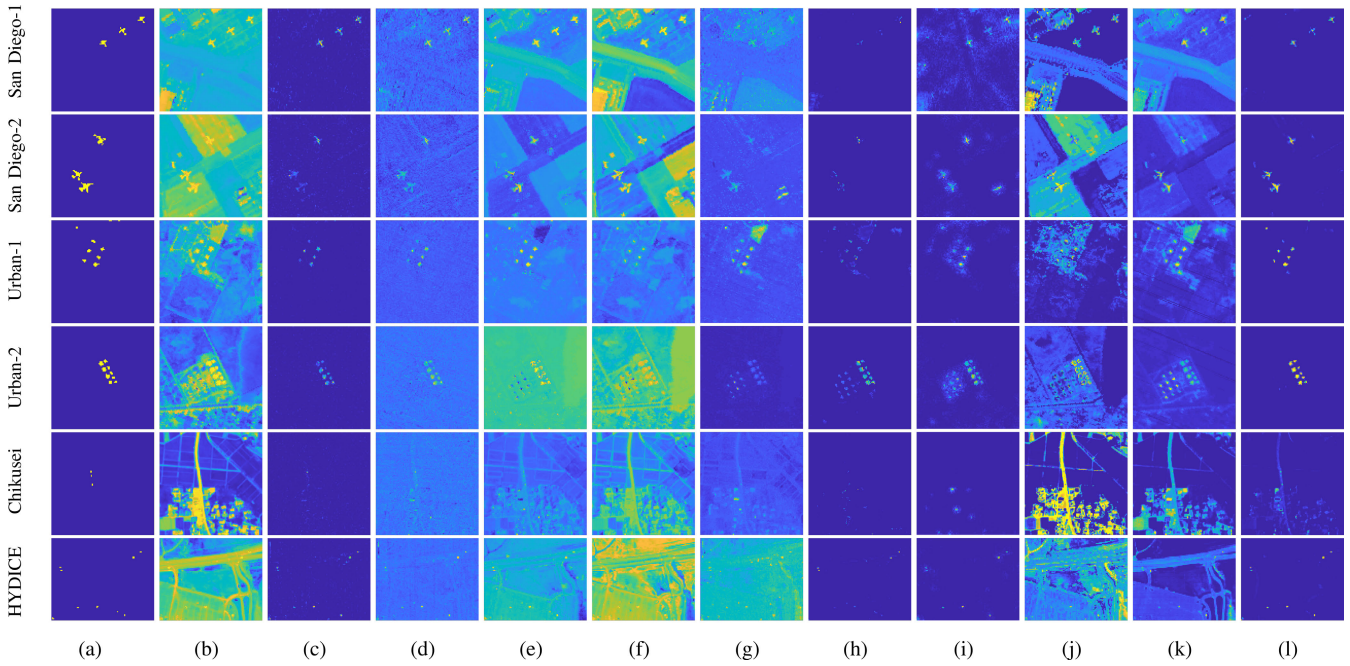


Fig. 12. Ground-truth images and detection maps in terms of comparison results of ten different detectors on six benchmark datasets. (a) Ground-truth. (b) SAM. (c) ACE. (d) CEM. (e) SRTD. (f) CSCR. (g) SSBSTD. (h) TSETD. (i) SSTD. (j) HTD-Net. (k) TSCNTD. (l) SASN.

corresponding to HTD-Net are the lowest, and its ROC curves are basically at the bottom right. Nevertheless, it performs better in target highlighting. In general, the three detectors have relatively good background suppression ability. Except for individual cases, their separation degree is also excellent since they take into account the separation of target and background in design.

Remarkably, for all six datasets, the overall AUC values of the proposed method are supreme and the ROC curves are at the top left on the whole. Although in some datasets, the ROC curves of some algorithms exceed the red curves corresponding to SASN at the beginning, with the increase of abscissa, the red curves always first reach the ordinate 1.0. In addition, the green boxes of SASN are infinitely close to the ordinate 0 and the distance between the two boxes is large, indicating preeminent background suppression ability and exceptional separation capacity. In conclusion, the proposed SASN not only has optimal detection performances but also has splendid generalization. Achieving such superior performance can be attributed to the success that the background training samples selected by MRBS are pure, multifarious, and representative. Meanwhile, the target training samples generated by TBRM do not mix other manual interference information, providing a reasonable and effective training set for network training. In addition, the three-branch network of SASN fully extracts spectral information and combines triplet-soft loss to promote spectral aggregation and separation.

2) Visual Evaluation: The detection maps of different algorithms on the six datasets are presented in Fig. 12. For SAM, it almost clearly highlights the targets, but for the last two datasets, the targets are confused in backgrounds. In addition, it is obvious that SAM has the worst ability to suppress backgrounds among the three traditional detectors. Moreover, it can be observed that ACE and CEM detect most of the targets, but CEM performs better in target highlighting. Specifically, ACE has a better effect on the background suppression of each dataset, while the background detection in CEM is greatly affected by noise. Except for the poor detection results of small targets in the Chikusei dataset with a complex background, SRTD, and CSCR obviously locate target pixels well, but their performances of background suppression are weak. SSBSTD is relatively neutral to target highlighting and background suppression, and the visual separation is not good for most datasets. TSETD achieves the best background suppression among these ML-based detectors, but unfortunately, its target detection capability is unconvincing. The DL- and ML-based detector SSTD has promising background suppression ability. Nevertheless, due to the guided filter, the target edges become blurred and the visual effect becomes worse. In addition, for the Chikusei dataset, SSTD hardly detects the targets. The DL-based algorithms HTD-Net and TSCNTD detect most target pixels on the whole, but their background vision is very messy, and they only suppress part of the background. Most prominently, the proposed SASN shows the best visual effect. For all six datasets, it can not only highlight the targets better but also has the superb ability to suppress the backgrounds, yielding an illustrious visual separation effect.

G. Analysis of Time Cost

Tables IV and V report the processing time of various detectors and the execution time of specific processes in DL-based detectors on the six datasets, respectively.

It can be obviously seen from Table IV that the three classic detectors SAM, ACE, and CEM take the shortest time, but they have poor competitiveness. Different execution times of SRTD and CSCR are required due to different inner and outer window settings in terms of the dual concentric window. In addition, when solving sparse vectors, it also

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TABLE IV
PROCESSING TIME (IN SECONDS) OF TEN DIFFERENT DETECTORS ON SIX BENCHMARK DATASETS

Time(s)	San Diego-1	San Diego-2	Urban-1	Urban-2	Chikusei	HYDICE
SAM	0.387	0.385	0.349	0.371	0.425	0.359
ACE	0.274	0.288	0.302	0.311	0.233	0.153
CEM	0.191	0.203	0.217	0.208	0.328	0.215
SRTD	20.373	8.091	5.318	5.298	11.255	4.179
CSCR	48.145	7.683	5.185	4.846	10.859	5.434
SSBSTD	407.761	598.704	370.344	458.344	1341.357	243.216
TSETD	15.276	14.861	16.639	16.511	46.899	11.835
SSTD	1305.337	1285.413	1250.759	1199.488	2539.514	978.339
HTD-Net	10444.398	10405.621	8751.715	9507.069	22046.377	7688.712
TSCNTD	1029.485	811.107	903.761	897.738	1505.613	765.077
SASN	430.022	443.596	467.3344	551.92	450.004	406.413

TABLE V
IMPLEMENT TIME (IN SECONDS) OF THREE DL-BASED DETECTORS ON SIX BENCHMARK DATASETS

Datasets	Process	HTD-Net	TSCNTD	**SASN**
San Diego-1	Preparation	9525.084	377.441	**40.778**
	Training	911.497	649.709	**386.521**
	Testing	7.817	**2.335**	2.723
San Diego-2	Preparation	9482.755	194.512	**18.286**
	Training	915.699	614.366	**422.492**
	Testing	7.167	**2.229**	2.818
Urban-1	Preparation	7883.229	285.982	**60.974**
	Training	861.639	615.352	**403.794**
	Testing	6.847	**2.427**	2.566
Urban-2	Preparation	8663.871	265.535	**133.142**
	Training	834.629	628.882	**416.239**
	Testing	8.569	3.321	**2.539**
Chikusei	Preparation	20612.178	868.351	**69.185**
	Training	1426.227	634.555	**377.865**
	Testing	7.972	**2.707**	2.954
HYDICE	Preparation	6770.199	117.791	**4.944**
	Training	911.356	645.057	**399.022**
	Testing	7.157	**2.229**	2.447

takes more time to use orthogonal matching pursuit (OMP). In particular, representation-based SSBSTD takes a longer time to solve the sparse vectors of the target and background, respectively. However, TSETD requires a shorter detection time. In particular, the last four algorithms take longer to execute because of the deep networks.

The training set preparation, training, and testing time of the last three DL-based detectors are given in Table V. Obviously, HTD-Net takes the longest time at each phase and has the lowest efficiency. It expends much time to generate the target samples by U-AE and gain copious training sets by pixel-pair subtraction, so there is no doubt that its preparation and training time are longer. In addition, the testing time of HTD-Net is longer due to the deeper network. The preparation of TSCNTD also requires OMP, which takes a longer time. The clustering process in preparation for SASN takes a long time, which is affected by parameter settings, while the time for background selection and target generation is extremely short. During training, both TSCNTD and SASN need to calculate the proposed loss functions, so the time is not short. Although TSCNTD has the shortest test time, SASN has the highest efficiency and the best detection performance as a whole.

V. CONCLUSION

To handle the problem of class imbalance and suppress backgrounds while highlighting targets simultaneously for HTD, we propose a SASN with TBRM. To be specific, for the background training set of SASN, we propose an MRBS incorporating DBSCAN and SAM to obtain a pure and representative background training set. To tackle the problem of class imbalance, TBRM is proposed to generate sufficient target training samples that are not mixed with manual interference information. Notably, TBRM is capable of simulating the spectral variability and forcing the network to extract global band information. After attaining the class-balanced training sets, with residual connection and SE attention mechanism, the SASN fully extracts high discriminative features and nonlinear ones in the spectra during training. Moreover, we develop a triplet-soft loss function in SASN to improve the separability. Specifically, the triplet-soft loss could make the network training proceed in the direction that not only the spectra of the target samples aggregate with that of the prior target but also the spectra of the background samples separate from that of the prior target and the target samples. During testing, the trained SASN simultaneously distinguishes the similarities and differences between the spectra of to-be-detected pixels and the prior target to highlight the targets and suppress the backgrounds. Compared with ten advanced HTD algorithms with competitive advantages, tons of experiments demonstrate the superior detection performances, background suppression ability, and separability of the proposed method on six experimental HSI datasets.

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